

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Permanent hardness can be removed by adding zeolite to hard water.

Reason (R): Zeolite decomposes and forms insoluble sodium zeolite.

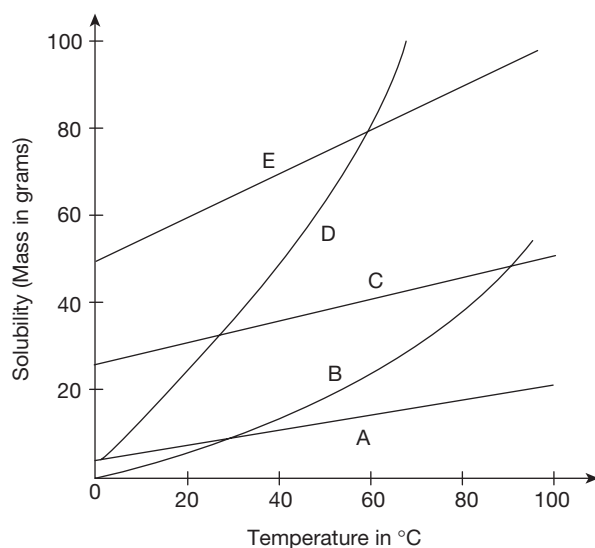
- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

2. **Assertion (A):** Graphite is a good conductor of electricity.

Reason (R): In graphite, carbon atoms are loosely bonded.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

3. Arrange the given solutes in the increasing order of their solubilities at 40°C.



- (a) EDBCA
- (b) ABCDE
- (c) CDBAE
- (d) BACDE

4. Certain metallurgical processes for the extraction of a particular metal are mentioned below. Arrange them in proper order.

- (A) Gravity separation
- (B) Smelting
- (C) Pulverization
- (D) Distillation
- (E) Roasting

- (a) ACEBD (b) CEADB
(c) AECDB (d) CAEBD

5. Match the statements of Column A with those of Column B.

Column A (Substances)	Column B (Properties)
(i) Wood charcoal	(A) Thick black substance
(ii) Sugar charcoal	(B) Brittle grey solid
(iii) Coke	(C) Light powdery black substance having a velvet touch
(iv) Animal charcoal	(D) Black porous solid
(v) Lamp black	(E) Less percentage of carbon

- (a) (i) → (C); (ii) → (A); (iii) → (E); (iv) → (B); (v) → (D)
 (b) (i) → (B); (ii) → (A); (iii) → (D); (iv) → (C); (v) → (E)
 (c) (i) → (B); (ii) → (D); (iii) → (A); (iv) → (E); (v) → (C)
 (d) (i) → (B); (ii) → (A); (iii) → (D); (iv) → (E); (v) → (C)

6. Which of the following ores can be separated by froth floatation?

- (a) Magnesite (b) Haematite
(c) Bauxite (d) Cinnabar

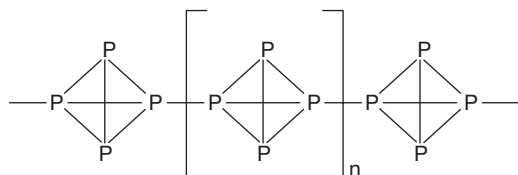
7. Identify the reaction(s) responsible for the drop in temperature in the middle region of the blast furnace.

- (a) $\text{CaCO}_3 \rightarrow \text{CO}_2 + \text{CaO}$ (b) $\text{CO}_2 + \text{C} \rightarrow \text{CO}$
 (c) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$ (d) Both (1) and (2)

8. A mixture of NH_3 and O_2 is passed over heated platinum. Identify one of the products of the reaction.

- (a) N_2 (b) N_2O (c) NO (d) NO_2

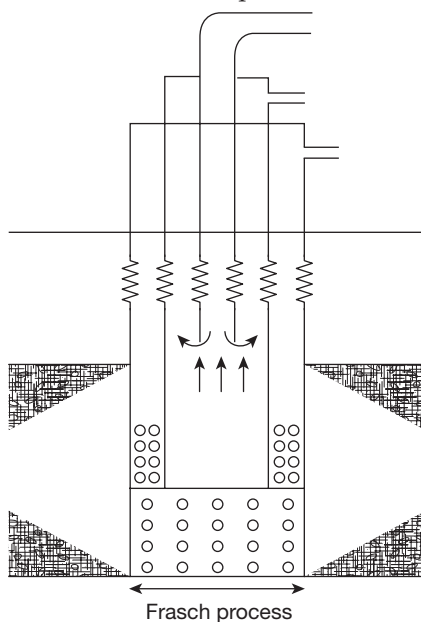
9. Which of the following uses of phosphorus is not related to the structure given below?



- (a) Making safety matches (b) Making strike anywhere matches
(c) Making fertilizers (d) Making phosphor bronze

Space for rough work

10. Identify the product obtained in the process shown below.



- (a) Molten sulphur (b) Sulphur vapour
(c) Flower of sulphur (d) Solid roll of sulphur
11. 1 kg of a saturated solution contains 200 g of solute at a given temperature. Calculate its solubility at that temperature.
(a) 20 (b) 25 (c) 50 (d) 10
12. The weight percentage of a solution is 10. Calculate the amount of solute to be added to 1 kg of the solution in order to saturate it. The solubility of the solute at that temperature is 25.
(a) 100 g (b) 25 g (c) 125 g (d) 25 g
13. The electrochemical equivalent of a divalent metal is x . Calculate the mass of the metal obtained at the cathode by the passage of 0.5 Faraday of electric charge.
(a) $0.5x$ g (b) $48,250x$ g
(c) $96,500x$ g (d) x g
14. If ' m ' g of a metal gets deposited by passing 1 A current for 1 s and the atomic weight of the metal is n , then calculate its valency assuming 1 Faraday = p Coulomb.
(a) $\frac{n \times p}{m}$ (b) $\frac{n}{m \times p}$ (c) $\frac{p}{m \times n}$ (d) $\frac{m \times p}{n}$

Space for rough work

15. Bleeding from a wound stops by the application of FeCl_3 solution. Which of the following phenomenon takes place due to the same reason?
- (a) Dust particles present in air become visible due to the passage of a narrow beam of light in a dark room.
 - (b) Some syrupy medicines are advised to be taken after shaking the bottle.
 - (c) A supersaturated solution slowly becomes transparent.
 - (d) Addition of alum hastens the process of purification of water.

Space for rough work

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Water is used in car radiators as a coolant.

Reason (R): Water has high specific heat capacity.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

2. **Assertion (A):** Diamond is a better conductor of heat than metals.

Reason (R): Diamond has no free electrons.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

3. Arrange the following electrolytes in the increasing order of amount of metal deposited at the cathode by passing same quantity of electricity.

- (A) NaCl (B) Al_2O_3
 (C) $\text{Fe}_2(\text{SO}_4)_3$ (D) ZnCl_2
 (a) CBDA (b) BCAD (c) BCDA (d) CBAD

4. Arrange the following ores in the order of the metals given below.

Metals: Calcium, iron, lead, manganese and thorium

- (A) Hematite (B) Pyrolusite
 (C) Monazite (D) Gypsum
 (E) Galena
 (a) DAEBC (b) DACBE
 (c) DABEC (d) DEACB

5. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Ammonia	(A) Used as rat poison
(ii) Phosphorus	(B) Manufacture of explosive such as TNT
(iii) Sulphur	(C) Purification of drinking water
(iv) Chlorine	(D) Vulcanization

- (a) (i) → (C); (ii) → (D); (iii) → (A); (iv) → (B)
 (b) (i) → (C); (ii) → (A); (iii) → (D); (iv) → (B)
 (c) (i) → (B); (ii) → (A); (iii) → (D); (iv) → (C)
 (d) (i) → (D); (ii) → (A); (iii) → (B); (iv) → (C)

6.10 Chapter 6 Water, Solutions, Solubility and Hydrogen; Metals and Non-metals

6. 180 g of 30% (w/w) NaOH is prepared. Calculate the volume of water utilised if the density of water is 0.9 g/cc.

(a) 130 mL (b) 140 mL (c) 125 mL (d) 135 mL

7. The amounts of solutes 'A', 'B' and 'C' in 300 g of water at 25°C, 40°C and 60°C in their saturated solutions are given below. Identify the true statement among the following:

	25°C	40°C	60°C
A	110 g	90 g	105 g
B	140 g	170 g	180 g
C	130 g	150 g	140 g

- (a) A, B and C form precipitates on cooling their respective saturated solutions at 60°C – 40°C.
(b) A and B form precipitates on cooling their respective saturated solutions from 40°C – 25°C.
(c) B and C form precipitates on cooling their respective saturated solutions from 60°C – 40°C.
(d) 'A' can form precipitate on cooling its saturated solution at 60°C – 40°C, whereas C forms precipitate on cooling its saturated solution at 40°C – 25°C.
8. Examine the constituents of the containers given below and identify the container in which change in colour is observed.

Container 'A': Glauber's salt + anhydrous copper sulphate

Container 'B': Hydrated MgCl_2 + anhydrous copper sulphate

Container 'C': Calcium oxide + anhydrous cobalt chloride

Container 'D': Anhydrous CaCl_2 + anhydrous cobalt chloride

- (a) Container C (b) Container D
(c) Container B (d) Container A
9. Observations made by students 'A', 'B', 'C' and 'D' are given below in the table. Identify whose report (or) observation is correct?

Student	Solution	Colloid	Suspension
A	Homogeneous	Brownian motion is significant.	Tyndall effect is significant.
B	Tyndall effect is observed.	Heterogeneous	Can be observed with naked eyes.
C	Particles never settle down.	Size of particles is 10 nm – 1 μm	Tyndall effect is insignificant.
D	Heterogeneous	Tyndall effect is observed.	Ordinary filtration

- (a) Student A (b) Student B
(c) Student C (d) Student D

Space for rough work

Space for rough work

10. Butter is a(an) _____.
(a) emulsion (b) sol
(c) gel (d) liquid aerosol
11. Which among the following is not used for making any article?
(a) Steel (b) Cast iron
(c) Pig iron (d) Wrought iron
12. Identify the true statement among the following:
(a) Red phosphorus is constituted of isolated tetrahedral units, whereas white phosphorus contains chains of tetrahedral units.
(b) Red phosphorus is constituted of chains of tetrahedral units, whereas white phosphorus contains isolated tetrahedral units.
(c) Both red and white phosphorus are constituted of isolated tetrahedral units.
(d) Both red and white phosphorus are constituted of chains of tetrahedral units.
13. Examine the following reaction: $\text{Cu} + \text{H}_2\text{SO}_4 \rightarrow \text{A} + \text{H}_2\text{O} + \text{SO}_2$
Aqueous solution of 'A' on treatment with H_2S forms 'B'. Identify A and B.
(a) CuSO_4 , CuS (b) Cu_2SO_4 , Cu_2S
(c) Cu_2SO_4 , CuS (d) CuSO_4 , Cu_2S
14. Which among the following pairs of compounds gives pale yellow precipitates with silver nitrate solution?
(a) KBr , NaCl (b) NaBr , NaCl
(c) NaCl , NaI (d) NaBr , NaI
15. The property exploited in the usage of argon in incandescent light bulbs along with mercury is _____.
(a) easy ionization
(b) low density
(c) prevention of oxidation of tungsten
(d) long life

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** The amount of heat energy required to raise the temperature of 3 g of water by 5°C is 42 Joule.

Reason (R): Specific heat capacity of water is $4.2 \text{ J/g}^{\circ}\text{C}$.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

2. **Assertion (A):** Graphite is a good conductor of electricity.

Reason (R): Graphite has free electrons.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

3. Arrange the following conditions in the decreasing order of the amount of magnesium obtained during the electrolysis of molten MgCl_2 .

- (a) $i = 5 \text{ A}$, $t = 15 \text{ min}$ (b) $i = 12 \text{ A}$, $t = 6 \text{ min}$
 (c) $i = 8 \text{ A}$, $t = 570 \text{ s}$ (d) $i = 14 \text{ A}$, $t = 330 \text{ s}$
 (a) CDBA (b) DCBA
 (c) BACD (d) CDAB

4. Arrange the following metals in the order of the ores given below.

Epsom salt, anglesite, limonite, cryolite and pitch blende

- (A) U (B) Al (C) Pb (D) Mg
 (E) Fe
 (a) DAEBC (b) DECAB
 (c) DCEBA (d) CDEBA

5. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Removal of grease	(A) Nitrogen
(ii) Storage of canned food	(B) Chlorine
(iii) Manufacture of gun powder	(C) Ammonia
(iv) Preparation of poisonous gases	(D) Sulphur

- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)
 (b) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B)
 (c) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)
 (d) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (C)

Space for rough work

6. Calculate the molarity of a g of $b\%$ (w/w) Ca(OH)_2 solution. Density of solution is x g/mL.

- (a) $0.135 ab$ M (b) $0.135 bx$ M
(c) $0.135 ax$ M (d) $0.135 abx$ M

7. The amounts of solutes 'A', 'B' and 'C' in 300 g of water at 25°C , 40°C and 60°C in their saturated solutions are given below. Identify the correct statement from the following when hot saturated solutions of A, B and C are cooled slowly from 60°C to 25°C .

	25°C	40°C	60°C
A	110 g	90 g	105 g
B	140 g	170 g	180 g
C	130 g	150 g	140 g

- (a) 5 g of A and 40 g of B are precipitated.
(b) 30 g of B and 10 g of C are precipitated.
(c) 15 g of A and 20 g of C are precipitated.
(d) 40 g of B and 10 g of C are precipitated.

8. Identify the true statement from the following.

- (a) All deliquescent substances are hydrated salts.
(b) All desiccating substances are hygroscopic substances.
(c) All anhydrous salts are deliquescent substances.
(d) All hydrated salts are efflorescent substances.

9. Addition of potash alum results in the purification of water. The phenomenon associated with this reaction is _____.

- (a) precipitation (b) crystallization
(c) coagulation (d) sterilization

10. In foam, dispersion medium is _____ and dispersed phase is _____, respectively.

- (a) liquid, gas (b) gas, liquid
(c) liquid, solid (d) solid, liquid

11. Among the following, identify the true statement.

- (a) Wrought iron is the purest form of iron and is more malleable.
(b) Wrought iron is the purest form of iron and is less malleable.
(c) Cast iron is the purest form of iron and is more malleable.
(d) Cast iron is the purest form of iron and is less malleable.

12. The electronic configuration of 'X' is $a, b, (b - a - 1)$. 'A' and 'B' are two allotropes of X and A is a soft solid; B sublimes at 565K . Identify the true statement regarding A and B.

- (a) $A \rightarrow \text{Red phosphorus}$; $B \rightarrow \text{White phosphorus}$
(b) $A \rightarrow \text{Crystalline allotrope}$; $B \rightarrow \text{Amorphous allotrope}$

- (c) Ignition temperature of A is greater than that of B.
 (d) A is more reactive than B.

13. The electronic configuration of metal 'X' is 2, 8, 13, 2. Aqueous solution of salt of X on treatment with H_2S forms a precipitate 'Y'. Identify X and the colour of the precipitate.
 (a) Pb, Flesh colour (b) Mn, White colour
 (c) Mn, Black colour (d) Mn, Flesh colour
14. Which among the following gives scarlet red precipitate with mercuric chloride solution?
 (a) HI (b) HF (c) HCl (d) HBr
15. Which among the following is used in advertisement sign boards?
 (a) Xenon (b) Argon (c) Helium (d) Neon

Space for rough work

Answer Keys

Assessment Test I

1. (d) 2. (a) 3. (b) 4. (a) 5. (b) 6. (a) 7. (b) 8. (a) 9. (a) 10. (b)
 11. (a) 12. (c) 13. (a) 14. (b) 15. (d)

Assessment Test II

1. (a) 2. (b) 3. (d) 4. (c) 5. (a) 6. (c) 7. (c) 8. (d) 9. (d) 10. (a)
 11. (a) 12. (b) 13. (d) 14. (d) 15. (d)

Assessment Test III

1. (d) 2. (c) 3. (a) 4. (b) 5. (b) 6. (c) 7. (a) 8. (a) 9. (a) 10. (c)
 11. (d) 12. (b) 13. (c) 14. (a) 15. (b)

Assessment Test IV

1. (c) 2. (d) 3. (c) 4. (c) 5. (a) 6. (d) 7. (c) 8. (d) 9. (c) 10. (c)
 11. (c) 12. (d) 13. (c) 14. (d) 15. (a)